

Ejer aut disco unidad traslada z1 0 z2

01

Ejercicio 1. Sean $z_1, z_2 \in \mathbb{D}$ tales que $z_1 \neq z_2$, demuestra que existe $f \in \text{Aut}(\mathbb{D})$ tal que $f(z_1) = 0$ y $f(z_2) \in (0, 1) \subset \mathbb{R}$.

Solución: Consideramos $g := f \circ \tau_{z_1}$, entonces $g \in \text{Aut}(\mathbb{D})$ y $g(0) = f(z_1) = 0$. Por la obs-aut-disco-unidad-fija-origen-imp-rotacion/Observación 1, $\exists \gamma \in \mathbb{T} : g = \rho_\gamma$. Por tanto, $f = g \circ \tau_{z_1}^{-1} = \rho_\gamma \circ \tau_{z_1}$.

Si imponemos ahora que $f(z_2) \in (0, 1)$, tenemos que $\rho_\gamma(\tau_{z_1}(z_2)) = \gamma \tau_{z_1}(z_2) \in (0, 1)$. Por tanto, $\gamma \tau_{z_1}(z_2) = |\tau_{z_1}(z_2)|$, luego $\gamma = \frac{|\tau_{z_1}(z_2)|}{\tau_{z_1}(z_2)} \in \mathbb{T}$. ■

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